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Insurance-Incentivized Wearable Biosurveillance for Cruise Ships: A Simulation-Based Economic Analysis

This report presents a simulation-based economic analysis of wearable biosurveillance for managing norovirus outbreaks on cruise ships. By employing the Crusher-to-the-Bridge epidemiological testbed, the study demonstrates that coupling early anomaly detection with an insurance-incentivized compliance framework yields significant public health benefits and a massive return on investment. The findings suggest that shifting from reactive clinical surveillance to proactive, technology-driven incentive models is a highly effective constraint on maritime disease transmission.

Overall Feedback

Here are some observations from reviewing the manuscript.

Baseline for the economic counterfactuals

The economic projections in Section 5.2 state a value creation modeled on 184 infections prevented at a 50% intervention effectiveness, which scales to 4.4 million dollars in annual savings. Readers cross-referencing Table 2 will note that clinical RDTs prevent 158 infections at that same effectiveness level, meaning the incremental value of adding wearables to the clinical baseline is 26 infections per 500 agents rather than 184. Additionally, the base economic case relies on the high-severity 46.9% attack-rate scenario, whereas Section 2.3 and Annex C establish dose_adjustment=7.0 with a 5-6% VSP median attack rate as the calibrated realistic baseline. Rebuilding the expected-value economics around precise counterfactual operations—comparing current practice, clinical-triggered redesign, and wearable-triggered redesign—and weighting those values against the empirical VSP severity distribution would clarify the specific financial advantage of the hardware.

Intervention modeling mechanisms

Section 4.1 establishes a critical insight: an identical operational protocol response produces identical outcomes regardless of the sensor triggering it. Section 4.2 then introduces an immediate 30-90% transmission reduction from the detection epoch onward. Readers will trace this back to the operational mechanics required to realize such a reduction, including standard operating procedures, staffing capacity, cabin-confinement compliance, food-service adjustments, and procedural decay. Because Section 6.1 recommends that wearable alerts trigger clinical follow-up rather than immediate confinement, the one-epoch detection advantage may yield a delayed protocol rollout in practice. Structurally modeling these delay algorithms and physical friction factors within the CTB framework, rather than applying an external transmission modifier, would anchor the effectiveness claims.

Wearable alert validation and false positives

The core case for this architecture rests on wearable anomaly detection firing at Day 0 with 100% coverage at a 1.4% attack rate. Practitioners deploying this system will look for the underlying algorithmic specifications: the precise alert threshold, baseline calculation period, positive predictive value, and false-alert burden. Section 8.3 notes that non-infectious causes like seasickness, alcohol consumption, and jet lag generate physiological anomalies. At a 4% baseline outbreak probability, incorporating these non-infectious signals and evaluating system performance across "fizzle" background cruises is necessary to establish the operational true-to-false alert ratio. Consequently, updating the economic model in Section 5 to account for medical staff time, confirmatory continuous RDT use, and passenger friction generated by false-alarm investigations will provide a more comprehensive cost profile.

Simulation parameter transparency

Section 2 and Annex A successfully map the CTB system inventory, repository structure, and epoch loop. Domain experts evaluating the outbreak dynamics will require deeper sightlines into the mechanical parameters driving the results, such as the assumed index-case counts, inter-agent contact schedules, route-specific transmission equations, stoplight algorithmic thresholds, and sampling cadences. Further, the methodology indicates that 50 outbreak-producing cruises per regime were selected from a pool of over 1,300 simulations. Detailing how this outbreak-conditioning step filters the simulation outputs will clarify how the Day 0 alerting and epoch 1 syndromic activation metrics might shift across the broader unconditioned sample.

Insurance mechanism enforceability

The structural solution to adoption in Section 7.3 hinges on conditional travel insurance coverage, where claims are honored only if the device remains active. Underwriters and operational planners will look for the contractual treatment of inevitable device downtime—charging intervals, aquatic activities, sleep behavior, hardware failures, missing ownership, age constraints, or specific medical exceptions. Further, Section 7.4 projects the travel insurer roughly breaking even, while Section 8.3 characterizes the large P&E premium benefit as illustrative rather than actuarially derived. Formalizing an explicit product design with claims eligibility logic, a defined loss-ratio model, and a concrete distribution of realized savings across passengers, lines, insurers, and P&E clubs will solidify the behavioral game theory framework.

Privacy and data governance constraints

Section 8.3 states that FDA classification, HIPAA, GDPR, and continuous physiological privacy considerations fall out of scope. Because the proposed intervention depends computationally and economically on streaming biometric data from thousands of leisure passengers, these governance choices function as foundational architecture variables rather than downstream compliance tasks. Specifying whether anomaly computations occur on-device, at the local network edge, or require transmitting raw biometric matrices to external servers shapes the infrastructure cost model. Defining the consent mechanisms, data retention limits, and dispute handling protocols directly informs the achievable opt-in rates and insurance enforceability analyzed elsewhere in the report.

Detailed Feedback

#1 Discrepancy between calibrated severity and main results

> *To ensure the simulation produced realistic outbreak dynamics, the pathogen doseadjustment parameter was calibrated against real-world CDC VSP data. The severity sweep across four dose_adjustment values {5.0, 6.0, 7.0, 8.0} identified dose_adjustment=7.0 as the optimal calibration point. At this severity, the simulation produced a median passenger attack rate of 6.4%, closely matching the 5.6% median observed in the historical VSP dataset of 204 norovirus outbreaks (1993-2025). The baseline 46.9% attack rate observed at higher severity (dose=6.0) was used to represent worst-case tail-risk events.*

There seems to be an issue with how the calibrated and high-severity scenarios are linked: the paper identifies `dose_adjustment=7.0` as the VSP-calibrated point, but the headline detection, null-result, and counterfactual analyses appear to rely on the 46.9% `dose=6.0` tail-risk baseline. This does not necessarily invalidate the stress-test analysis, but the paper should distinguish more clearly which results are calibrated-median findings and which are high-severity tail-risk findings.

#2 Party count and value allocation in Section 7.4

> ### *7.4 Three-Party Co-Optimization*
>
> *Figure 7. Co-optimization value flow: annual costs and benefits across four parties under insurance-incentivized wearable surveillance (200 -cruise fleet, 3,000 passengers/cruise, 70% BYOD adoption).*
>
> *The insurance incentive framework creates value for all three parties simultaneously:*
>
> *Passenger: - Cost: $0 (brings existing smartwatch) - Benefit: \sim \ 4.65 /$ cruise (insurance discount + reduced illness risk) - Mechanism: Conditional insurance coverage ensures device stays active*
>
> *Cruise Line: - Cost: \ 400 \mathrm{~K} /year*$(analyticsplatformfor200 - cruisefleet) - Benefit:$ *$ 3.2 \mathrm{M} /year*$(reducedoutbreakcosts)$*+$ 4.7 \mathrm{M} / year*$(P\&Ipremiumreduction)Totalnetbenefit:$*\sim $ 7.9 \mathrm{M} /$ year - Mechanism: P&I clubs (maritime mutual insurers) price outbreak risk based on loss history; demonstrated surveillance adoption is provable risk reduction, analogous to ship classification societies tiering premiums by installed safety systems*
>
> *Travel Insurer: - Cost: \sim \ 167 \mathrm{~K} /year*$(discountsofferedtoparticipants) - Benefit:$ *\sim $ 167 \mathrm{~K} /$ year (reduced claims from lower outbreak incidence) - Additional value: Customer retention, competitive differentiation, self-selection effect (healthconscious travelers who opt in are lower-risk policyholders - the same adverseselection advantage observed in auto telematics)*
>
> *P&I Club (Maritime Mutual Insurer): - Cost: $0 (reduces premium, no new expenditure) - Benefit: Lower loss ratios across the mutual pool - Mechanism: Experience-rated premium adjustment based on demonstrated risk reduction.*

Section 7.4's value-flow accounting is internally inconsistent: the title and lead sentence say three parties, but the caption and bullets identify four. In addition, the $4.7M P&I premium reduction is assigned to the cruise line in the text while Figure 7 displays a $4.7M benefit for the P&I Club. Specifically, Figure 7 shows the Cruise Line with a $3.2M benefit, $0.4M cost, and $2.8M net, whereas the text's stated cruise-line "net" of about $7.9M appears to sum $3.2M + $4.7M without subtracting the $400K platform cost. The figure and text should be reconciled as to whether the $4.7M is a cruise-line premium saving, a P&I loss-ratio benefit, or both under a clearly defined transfer accounting.

#3 Flawed expected utility calculations for insurance compliance

> *With conditional insurance coverage: - EU(comply) =+\ 1,125(-\$ 375* $confinement$*+\$ 1,500* $expected insurance value if ill) - EU(evade)$ *=-\$ 1,262\$ (lose insurance coverage entirely + detection risk + illness risk)*

The conditional-insurance payoff matrix mixes insurance coverage value and illness losses in a way that makes the payoff baseline unclear. The $1,125 compliance payoff treats $1,500 of insurance value as a positive payoff, even though travel insurance would ordinarily indemnify illness-related losses rather than create a stand-alone gain, and the $1,262 evasion loss includes an additional lost-coverage component beyond the explicitly listed detection and illness risks. Specifically, the stated $1,262 evasion loss can be recovered only if the lost-coverage component is about $900 (apparently 30% × $3,000 from Annex B). Because the probability basis for these insurance-value terms is not stated consistently, and without specifying whether insurance value represents expected indemnified losses, risk-protection utility, or a transfer, the $2,387 compliance differential is hard to interpret.

#4 Wastewater timing is missing from modality results

> *Five surveillance regimes were evaluated, with 50 outbreak-producing cruises per regime (from a larger pool of 1,300+ total simulations):*
>
> 1. *Status Quo: Clinical RDT and syndromic surveillance only (baseline).*
> 2.
> ```
>  + Wearable: Status Quo plus 100\% adoption of continuous physiological monitoring.
> ```
> 3.
> ```
>  + Wastewater: Status Quo plus metagenomic wastewater sequencing in main effluent
> zones.
> ```
> 4. *Full Stack: All modalities enabled.*
> 5. *BYOD 70%: Full Stack, but wearable participation restricted to a realistic 70% "Bring Your Own Device" voluntary adoption rate.*

Wastewater sequencing is a named surveillance regime, but its quantitative timing is not reported alongside the other modalities in Table 1 or Table 2. Although the later discussion says wastewater does not improve time-to-detection and is better treated as a pathogen-identification layer, the results would be easier to interpret if the wastewater signal's detection epoch, attack rate at availability, coverage, or counterfactual contribution were shown or explicitly bounded.

#5 Growth-rate statement overstates the table

> *The mean epidemic curve across 50 simulated norovirus outbreaks reveals explosive early growth:*
>
> | *Epoch* | *Mean AR* | *Growth Ratio* |
> | --- | --- | --- |
> | *0* | *1.3%* | – |
> | *1* | *4.0%* | $3.05\times$ |
> | *2* | *6.7%* | $1.68\times$ |
> | *3* | *11.4%* | *1.69×* |
> | *5* | *22.6%* | – |
> | *7* | *33.3%* | – |
> | *10* | *42.5%* | – |
> | *13 (end)* | *46.9%* | $1.02\times$ |
>
> *The* $3\times$ *growth ratio in the first epoch means that every day of detection delay costs approximately three times as many infections.*

The statement that "every day" of detection delay costs approximately three times as many infections overgeneralizes from the first epoch. The table supports an approximately 3× increase only from epoch 0 to epoch 1; later ratios are much lower and the curve approaches saturation. Furthermore, the counterfactual results also show diminishing marginal gains from progressively later detection, rather than a constant exponential penalty.